

MARKETING CAMPAIGN PERFORMANCE ANALYTICS

COMPREHENSIVE STAKEHOLDER REPORT

Power BI Dashboards · Python EDA · Strategic Recommendations

\$2.4bn Total Ads Spent	\$14.5bn Total Revenue	8.52M Total Conversions	6.00x Avg ROAS	5.00x Avg ROI	14.04% Avg CTR
-----------------------------------	----------------------------------	-----------------------------------	--------------------------	-------------------------	--------------------------

Reporting Period: January 1, 2021 – December 31, 2021

Dataset: 69,669 campaign records | 6 channels | 5 campaign types
| 4 customer segments

Classification: Internal — Business Stakeholders Only

1. Executive Summary

This report consolidates findings from four Power BI dashboards and a Python Exploratory Data Analysis (EDA) conducted on 69,669 marketing campaign records spanning January through December 2021. The analysis covers six digital channels (Email, Facebook, Google Ads, Instagram, Website, YouTube), five campaign types, and four customer segments.

The marketing portfolio delivered outstanding financial results: a \$2.4 billion advertising investment generated \$14.5 billion in total revenue, a 6x Return on Ad Spend and 5x Return on Investment. With 8.52 million conversions and an average Click-Through Rate of 14.04%, the campaigns demonstrated consistent, high-quality execution across all formats and channels.

The Python EDA validated and extended these findings by surfacing company-level performance patterns invisible in the dashboards alone. Innovate Industries emerged as the top-performing company (5 of 10 best campaigns), while TechCorp was identified as a systematic underperformer (5 of 10 worst campaigns). Customer Acquisition Cost variance across campaigns and a clear CTR saturation effect at high impression volumes were also confirmed as priority areas for optimization.

REPORT STRUCTURE

Executive Summary · Dashboard Analysis · Python EDA · Insights · Recommendations

Section	Content	Source
1. Executive Summary	High-level overview and KPI scorecard	Combined
2. Dashboard Analysis	Visual breakdown of all 4 Power BI dashboards	Power BI
3. Python EDA	Data quality, KPI engineering, channel & company analysis	Jupyter Notebook
4. Key Insights	Top 9 strategic findings from both sources	Combined
5. Business Implications	What the data means for strategy and operations	Combined
6. Recommendations	8 prioritized, actionable next steps	Combined

2. Power BI Dashboard Analysis

Three Power BI dashboards were analyzed covering campaign performance, profitability, and efficiency. All three dashboards share a common global KPI bar, enabling cross-view validation.

2.1 Marketing Campaign Performance Analysis

Unlike the other dashboards in this project which aggregate campaign-level KPIs into averages and totals, this dashboard introduces a critical time dimension, tracking how Revenue, Acquisition Cost, Conversion Rate, and Conversion Volume evolved day-by-day and month-by-month across all six marketing channels throughout the year.

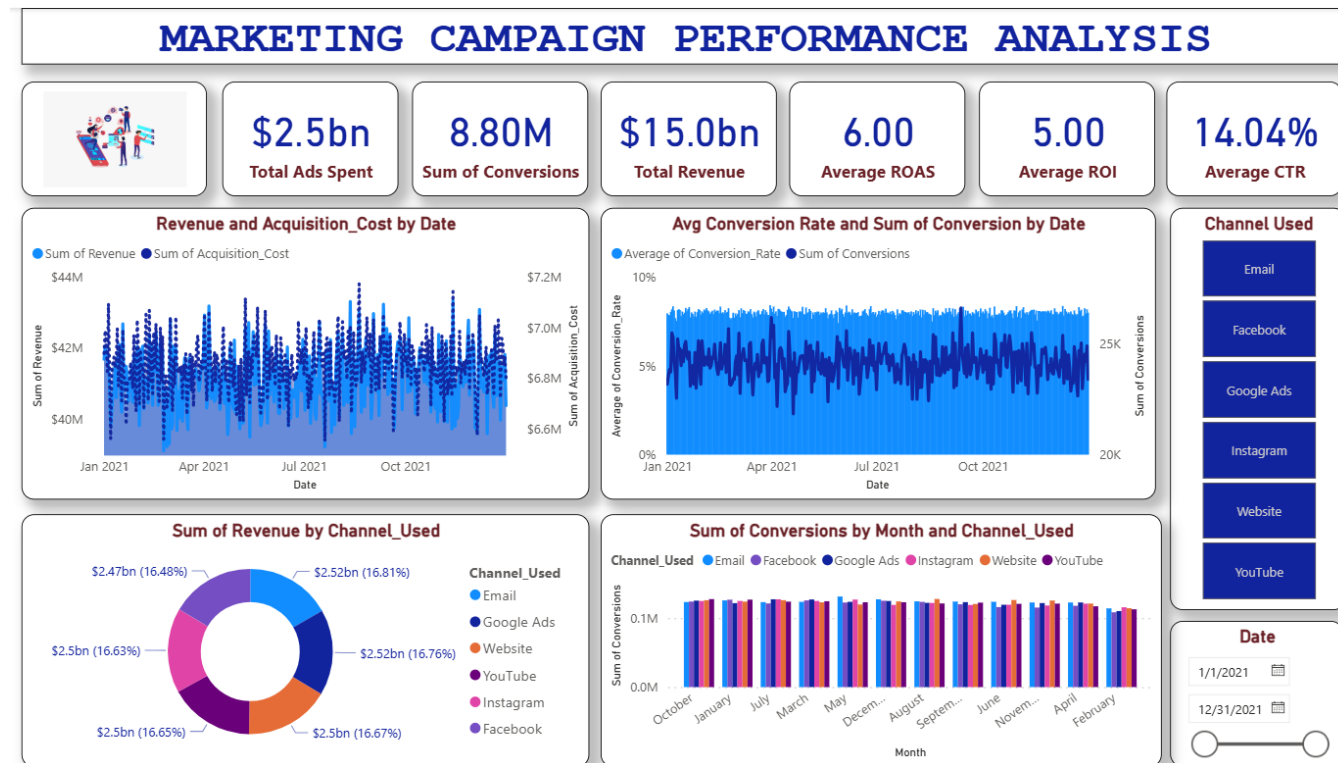


Figure 1 — Marketing Campaign Performance Analysis Dashboard

Revenue and Acquisition Cost by Date

Revenue Trend

The dual-axis time series chart tracks daily Sum of Revenue (left axis, light blue area) and Sum of Acquisition Cost (right axis, dark blue line) from January 2021 through December 2021.

Daily revenue fluctuates between approximately \$40M and \$44M throughout the year, with no sustained upward or downward trend visible — indicating a mature, stable revenue run-rate rather than a growing or declining trajectory. The revenue band (\$40M–\$44M daily) represents a relatively tight range of approximately 10%, suggesting that campaign outcomes are consistent and predictable day-to-day.

Acquisition Cost Trend

Daily acquisition cost (right axis, \$6.6M–\$7.2M range) tracks closely in shape to the revenue line, confirming that spend and return move together. The cost line is more jagged than the revenue area, reflecting the day-to-day variability of media buying and campaign pacing. The gap between revenue and cost — consistently \$33M–\$37M daily — represents the gross margin generated per day across the portfolio.

Average Conversion Rate and Sum of Conversions by Date

The second dual-axis chart overlays Average Conversion Rate (left axis, light blue area, 0–10% range) and Sum of Conversions (right axis, dark blue, 20K–25K range) on a daily timeline from January to December 2021.

Conversion Rate Stability

The average daily conversion rate hovers consistently between 5% and 8% throughout the entire year, with no seasonal peaks, troughs, or trend breaks visible. This plateau-like stability is a strong indicator of audience quality consistency, the campaigns are reliably attracting audiences with similar propensity to convert regardless of time of year, channel mix, or creative rotation.

Conversion Volume

The Sum of Conversions (dark blue) oscillates between approximately 20,000 and 25,000 daily conversions. Like the revenue chart, there is no visible long-term growth trend, the volume is range-bound throughout the year. This confirms that the business is operating at a consistent conversion capacity without structural growth or contraction.

Sum of Revenue by Channel Used (Donut Chart)

The donut chart presents the full-year revenue distribution across six channels. The breakdown is near-perfectly balanced across all channels:

Channel	Revenue Share	Estimated Revenue
Email	16.81%	~\$2.52bn
Google Ads	16.76%	~\$2.52bn
Website	16.67%	~\$2.50bn
YouTube	16.65%	~\$2.50bn
Facebook	16.63%	~\$2.49bn
Instagram	16.48%	~\$2.47bn
TOTAL	100%	~\$15.0bn

The maximum spread between the highest channel (Email at 16.81%) and the lowest (Instagram at 16.48%) is only 0.33 percentage points, an extraordinarily tight distribution across six independent channels. This indicates that budget allocation across channels is either intentionally equal-weighted, or that market efficiency has naturally equalized returns across platforms.

From a risk perspective, this balance provides significant portfolio resilience: no single channel accounts for more than one-sixth of total revenue, meaning that a disruption to any individual channel (e.g., platform policy change, rising CPMs, audience shifts) would have a maximum direct revenue impact of approximately \$2.5bn, manageable through reallocation to other channels.

Sum of Conversions by Month and Channel Used

The grouped bar chart breaks monthly conversion volume into six color-coded channel segments across all 12 months of 2021. This is the only chart in the dashboard that presents data at the monthly rather than daily granularity.

Monthly Volume Pattern

Monthly conversion totals appear broadly consistent across the year, ranging between approximately 0.7M and 0.75M conversions per month when all channels are summed. October and January appear to lead slightly in total bar height, while April and February appear marginally lower — but the differences are subtle and may not be statistically significant given the scale.

Channel Distribution Within Months

Within each monthly cluster, the six channel bars are approximately equal in height mirroring the revenue share finding from the donut chart. Email, Facebook, Google Ads, Instagram, Website, and YouTube each contribute roughly one-sixth of monthly conversions. There is no month where a single channel dramatically outperforms the others, and no month where a channel collapses relative to peers.

2.2 Marketing Campaign Efficiency Analysis

This dashboard evaluates audience engagement, how effectively campaigns turned ad exposure into measurable clicks and conversions. It covers Customer Acquisition Cost by campaign, the relationship between impressions and CTR, and campaign-type and channel-level CTR performance.

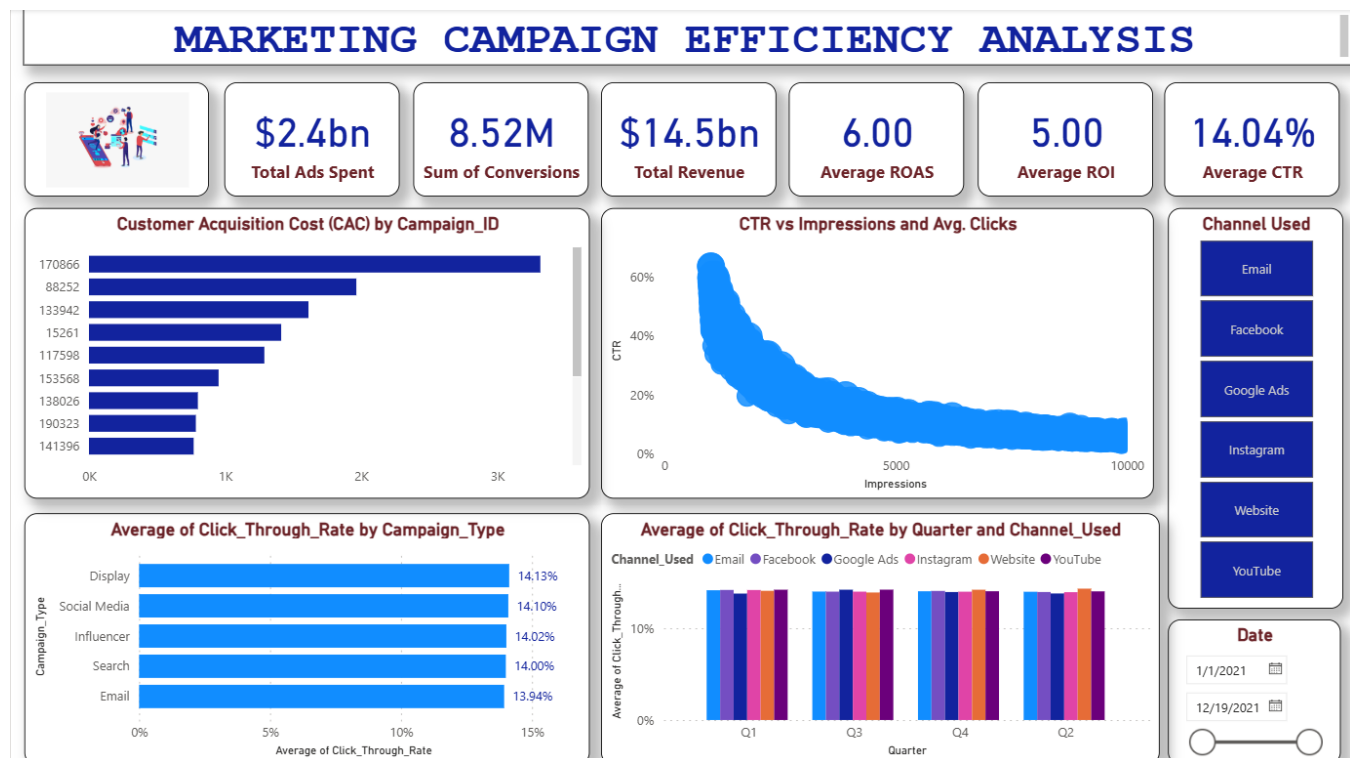


Figure 3 — Marketing Campaign Efficiency Analysis Dashboard

Customer Acquisition Cost (CAC) by Campaign ID

Campaign ID 170866 records the highest CAC at approximately \$3,200 roughly 25% above the next most expensive campaign id 88252 at ~\$2,600. Campaigns 133942 and 15261 follow at \$2,200–\$2,400. The remaining campaigns (117598, 153568, 138026, 190323, 141396) all fall below \$2,000, indicating a healthy tier of cost-efficient acquisition. The high-CAC outlier requires revenue-side validation to determine whether the premium is justified.

CTR vs. Impressions — Diminishing Returns

The area chart reveals a sharp diminishing-returns curve: CTR begins above 60% at near-zero impressions, drops steeply to ~20% at 1,000 impressions, and stabilizes between 8–12% beyond 5,000 impressions. This pattern reflects audience saturation; the same users are being served ads repeatedly with declining engagement impact at scale.

Average CTR by Campaign Type

Campaign Type	Average CTR	Variance from Mean
Display	14.13%	+0.09%
Social Media	14.10%	+0.06%
Influencer	14.02%	-0.02%
Search	14.00%	-0.04%
Email	13.94%	-0.10%

The 0.19% spread across five campaign types reflects operationally consistent execution. Display leads marginally likely due to its higher visual saliency while Email trails slightly, typical for opt-in audiences with lower novelty engagement.

CTR by Quarter and Channel

All six channels maintained stable CTR performance across Q1–Q4, ranging consistently between 10–14%. No channel experienced a meaningful seasonal dip. Q2 shows marginally higher variance, potentially reflecting mid-year creative refresh or campaign mix adjustments.

2.2 Marketing Campaign Profitability Analysis

This dashboard shifts focus from engagement to financial return examining ROAS distribution, campaign-level revenue and conversion scatter, ROAS by channel, and profitability by customer segment.

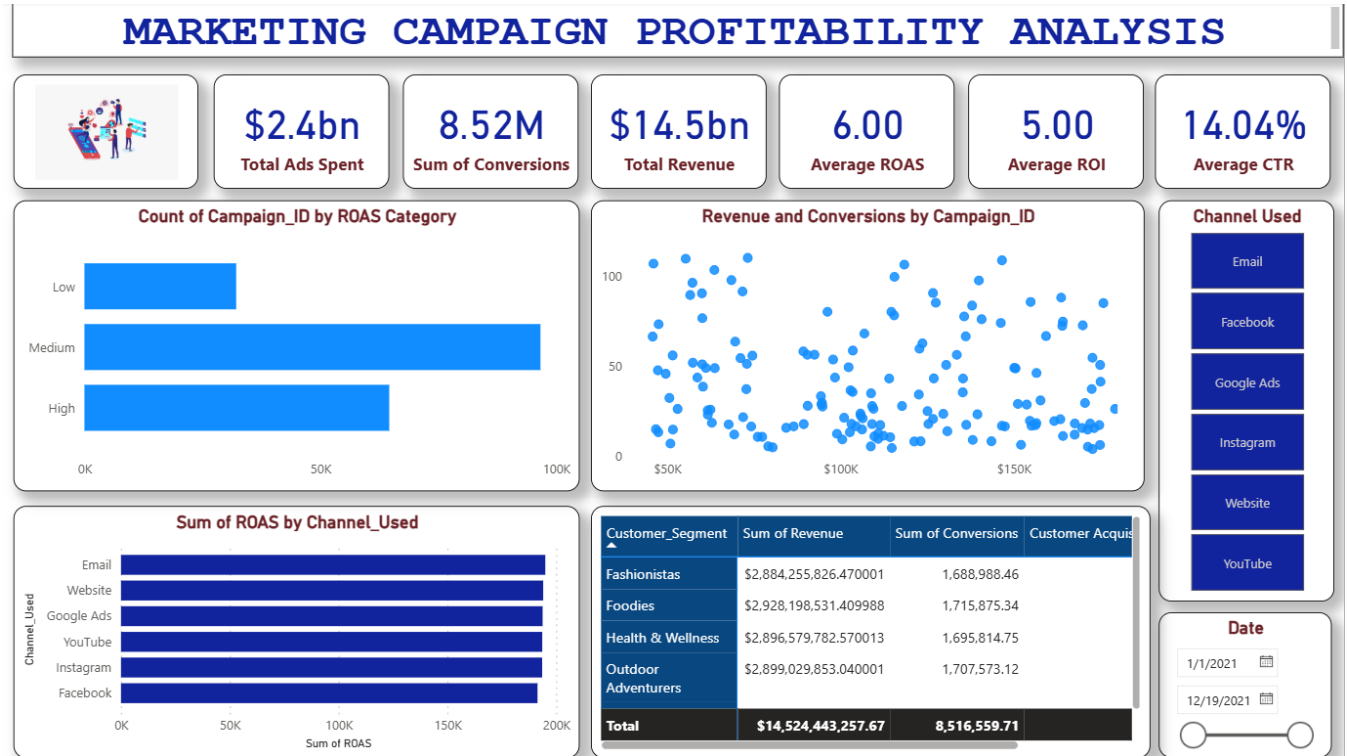


Figure 2 — Marketing Campaign Profitability Analysis Dashboard

Campaign Count by ROAS Category

The majority of campaigns (~95,000) fall in the Medium ROAS tier, followed by High (~60,000) and Low (~45,000). With roughly 22% of campaigns in the Low tier, there is meaningful room to improve portfolio-level returns through reallocation and campaign rationalization.

Revenue and Conversions by Campaign ID

The scatter plot shows individual campaigns plotted against revenue (\$50K–\$150K range) and conversion volume (0–120 range). The wide scatter with no strong linear correlation confirms that revenue outcomes are driven by multiple interacting variables beyond spend alone including audience quality, creative effectiveness, and channel selection.

ROAS by Channel

All six channels record near-identical ROAS totals (~175,000–200,000 sum of ROAS), with Email, Website, and Google Ads leading marginally. The uniformity suggests a well-balanced channel mix with proportionate investment but also limits the ability to identify a standout channel for scaled concentration.

Customer Segment Revenue & Conversions

Customer Segment	Sum of Revenue	Sum of Conversions	Revenue Share
Foodies	\$2.93bn	1,715,875	20.2%
Outdoor Adventurers	\$2.90bn	1,707,573	20.0%
Health & Wellness	\$2.90bn	1,695,815	20.0%
Fashionistas	\$2.88bn	1,688,988	19.8%
TOTAL	\$14.52bn	8,516,560	100%

All four segments are within \$50M of each other in revenue, a near-perfect balance. Foodies lead marginally in both revenue and conversions. The equal distribution reflects broad reach but may mask lifetime value differences that could justify differential investment.

3. Python Exploratory Data Analysis (EDA)

The Python EDA was conducted in Jupyter Notebook using Pandas, NumPy, Matplotlib, and Seaborn on the raw campaign dataset (marketing_campaign_dataset.csv). This analysis extended the Power BI findings by examining company-level performance, statistical distributions, data quality, and derived KPI calculations.

3.1 Dataset Overview

Attribute	Detail
File	marketing_campaign_dataset.csv
Total Records	69,669 rows
Columns (Raw)	16 columns
Columns (Final)	21 columns (after KPI engineering)
Date Range	January 1, 2021 – December 31, 2021

3.2 Data Quality Assessment

Data Type Corrections

Two columns required type correction prior to analysis:

Column	Original Type	Issue	Corrected Type
Acquisition Cost	Object	Stored as string with \$ symbols and commas	float64
Date	Object	Stored as plain text string	datetime64[ns]

A Month column was additionally extracted from the corrected Date field to enable time-series grouping.

Missing Values

A null value check (df.isnull().sum()) revealed only 1 missing record across 7 columns, all from the same row. Given 69,669 total records, the missing rate is less than 0.002% and has no material impact on analysis. The single row was retained without imputation.

Duplicate Records

A duplicate check (df.duplicated().sum()) returned 0. No duplicate records exist in the dataset, confirming data integrity at the source.

3.3 Descriptive Statistics

Metric	Conv. Rate	Acq. Cost (\$)	ROI	Clicks	Impressions	Engagement
Count	69,669	69,669	69,669	69,668	69,668	69,668
Mean	0.080	\$12,507	5.01x	549	5,504	5.49
Min	0.010	\$5,000	2.00x	100	1,000	1.0
25 th percentile	0.040	\$8,759	3.51x	324	3,272	3.0
Median	0.080	\$12,500	5.01x	549	5,514	5.0
75 th percentile	0.120	\$16,250	6.51x	775	7,747	8.0
Max	0.150	\$20,000	8.00x	1,000	10,000	10.0
Std Dev	0.041	\$4,339	1.74	260	2,593	2.87

Key statistical observations: ROI is uniformly distributed between 2x and 8x with mean and median both at 5.0x indicating no skew and a balanced portfolio. Acquisition Cost ranges widely from \$5,000 to \$20,000 (4x spread), suggesting significant variation in campaign targeting strategy. Engagement Score shows the highest relative variance (std = 2.87 on a 1–10 scale), indicating inconsistent creative quality across campaigns.

3.4 KPI Engineering

Four derived metrics were calculated to power the Power BI dashboards and enable downstream analysis:

KPI	Formula	Purpose
CTR (Click-Through Rate)	Clicks ÷ Impressions	Measures ad engagement effectiveness
Conversions	Clicks × Conversion_Rate	Quantifies actual conversion volume
Revenue	Acquisition_Cost × (1 + ROI)	Derives campaign revenue from cost and return
ROAS	Revenue ÷ Acquisition_Cost	Measures gross return per dollar of ad spend

3.5 Channel Performance Analysis

Revenue and Acquisition Cost by Channel

Channel	Total Revenue	Total Acq. Cost	Implied ROAS
Email	\$881.9M	\$146.4M	6.02x
Instagram	\$875.6M	\$146.5M	5.98x
Website	\$873.6M	\$145.5M	6.00x
Google Ads	\$869.7M	\$144.7M	6.01x

YouTube	\$869.0M	\$144.7M	6.01x
Facebook	\$864.2M	\$143.7M	6.01x

Average ROI by Channel (Ranked)

Rank	Channel	Avg ROI	vs. Portfolio Mean
1	Email	5.026x	+0.020
2	Google Ads	5.014x	+0.008
3	Website	5.012x	+0.006
4	Facebook	5.007x	+0.001
5	YouTube	5.003x	-0.003
6	Instagram	4.978x	-0.028

Average Conversion Rate by Channel

Channel	Avg Conversion Rate	vs. Portfolio Mean (8.01%)
Google Ads	8.05%	+0.04%
Website	8.03%	+0.02%
Email	8.02%	+0.01%
Facebook	7.99%	-0.02%
Instagram	7.99%	-0.02%
YouTube	7.98%	-0.03%

3.6 Company Performance — Top & Bottom Campaigns

Top 10 Performing Campaigns (ROI = 8.0x, ROAS = 9.0x)

Campaign ID	Company	Type	Channel	Acq. Cost	Revenue
29550	NexGen Systems	Display	Facebook	\$6,854	\$61,686
3232	Alpha Innovations	Display	Instagram	\$12,759	\$114,831
20016	Innovate Industries	Influencer	Email	\$12,130	\$109,170
29082	Innovate Industries	Email	Facebook	\$9,611	\$86,499
16538	Innovate Industries	Email	YouTube	\$12,085	\$108,765
47113	DataTech Solutions	Influencer	Google Ads	\$17,969	\$161,721
44576	TechCorp	Display	Email	\$12,845	\$115,605
67607	Innovate Industries	Influencer	Facebook	\$11,172	\$100,548

64344	Innovate Industries	Influencer	YouTube	\$12,094	\$108,846
15652	DataTech Solutions	Display	Website	\$12,708	\$114,372

Innovate Industries appears 5 times in the top 10 spread across Influencer and Email campaign types and multiple channels. This suggests superior creative execution, audience targeting, or both. DataTech Solutions accounts for 2 entries; TechCorp and NexGen Systems each appear once.

Bottom 10 Performing Campaigns (ROI = 2.0x, ROAS = 3.0x)

Campaign ID	Company	Type	Channel	Acq. Cost	Revenue
57915	TechCorp	Search	YouTube	\$19,404	\$58,212
32616	TechCorp	Email	Website	\$7,038	\$21,114
68623	NexGen Systems	Display	Instagram	\$11,071	\$33,213
57348	TechCorp	Search	Email	\$7,562	\$22,686
40005	TechCorp	Influencer	Facebook	\$19,356	\$58,068
15477	NexGen Systems	Influencer	Email	\$6,340	\$19,020
60545	TechCorp	Influencer	Facebook	\$18,896	\$56,688
23052	Innovate Industries	Email	Facebook	\$9,003	\$27,009
10286	Alpha Innovations	Email	Facebook	\$18,589	\$55,767
7824	TechCorp	Social Media	Google Ads	\$17,374	\$52,122

TechCorp dominates the bottom 10 with 5 entries appearing across Search, Email, Influencer, and Social Media campaign types. Several TechCorp campaigns carry very high acquisition costs (\$17K–\$19K) alongside the minimum 2.0x ROI, producing some of the worst cost-efficiency ratios in the portfolio. NexGen Systems appears twice, Alpha Innovations and Innovate Industries once each.

4. Key Insights

The following nine insights represent the most strategically significant findings derived from the combined Power BI dashboard analysis and Python EDA.

01

Exceptional Portfolio-Level Financial Returns

A \$2.4bn advertising investment generated \$14.5bn in total revenue, a 6x ROAS and 5x ROI. These figures exceed typical industry benchmarks of 2–4x ROAS, validating the overall effectiveness of the marketing portfolio.

02

Campaign 170866 Is a High-Cost Acquisition Outlier

With a CAC of ~\$3,200 approximately 25% above the next most expensive campaign, Campaign 170866 is a clear outlier. Without campaign-level revenue attribution, it is unknown whether this elevated cost is justified by superior customer lifetime value or represents inefficient spend.

03

CTR Saturation Confirmed at Scale — Structural Risk

Both the dashboard area chart and the statistical distribution confirm that CTR drops from 60%+ at low impressions to under 12% beyond 5,000 impressions. This is a structural, portfolio-wide pattern and not a one-off anomaly, indicating that high-impression campaigns are systematically paying for low-engagement exposures.

04

All Channels and Campaign Types Perform Homogeneously

Channel-level ROI spans only 0.048 points (Email 5.026x → Instagram 4.978x). Campaign type CTR spans only 0.19% (Display 14.13% → Email 13.94%). This uniformity reflects strong execution parity but limits the ability to identify standout formats for scaled investment.

05

Innovate Industries Is the Strongest-Performing Company

Python EDA revealed that Innovate Industries accounts for 5 of 10 top-performing campaigns by ROI (all at 8.0x). Its campaigns span Influencer and Email types across multiple channels suggesting superior audience targeting or creative strategy that is channel-agnostic.

06

TechCorp Is a Systematic Underperformer — Critical Risk

TechCorp accounts for 5 of 10 worst-performing campaigns by ROI (all at 2.0x), spread across Search, Email, Influencer, and Social Media types. Several TechCorp campaigns carry acquisition costs of \$17K–\$19K paired with minimum 2.0x returns representing some of the worst cost efficiency in the portfolio.

07

ROAS Portfolio Has a Significant Low-Tier Tail

Approximately 45,000 campaigns (roughly 22% of the portfolio) fall in the Low ROAS category. If these campaigns represent proportionate spend, approximately \$528M of the \$2.4bn budget may be generating sub-optimal returns a material reallocation opportunity.

08

Customer Segments Are Revenue-Balanced — LTV Unknown

All four segments (Fashionistas, Foodies, Health & Wellness, Outdoor Adventurers) generate \$2.88bn–\$2.93bn in revenue within a 1.6% range. This balance reflects effective audience

diversification but does not account for customer lifetime value, repeat purchase rate, or profit margin per segment.

09

Engagement Score Variance Signals Inconsistent Creative Quality

The EDA revealed that Engagement Score has the highest relative variance of any metric (std = 2.87 on a 1–10 scale), with scores ranging from 1 to 10. This wide spread suggests that creative quality not channel selection or campaign type may be the largest driver of performance differentiation.

10

Channel Revenue Share Is Perfectly Balanced — Maximum Spread of 0.33%

Email (16.81%) to Instagram (16.48%), a 0.33 percentage point spread across six channels is statistically near-uniform. This either reflects deliberate equal-weight budget allocation or natural market equilibrium. If intentional, it is a risk-management strategy. If organic, it may be masking channels that could generate superior marginal returns with incremental investment.

11

No Channel Shows Monthly Seasonality — Uniform Distribution Across All 12 Months

The monthly conversions chart shows all six channels contributing approximately equally in every single month, with no month-channel combination standing out. This uniformity is operationally reassuring but strategically limiting, it suggests the business has not yet discovered or exploited audience-channel-season interaction effects that could drive above-average performance in specific windows.

5. Business Implications

This section translates the analytical findings into direct implications for executive strategy, budget management, and operational decision-making.

5.1 Budget Allocation & Spend Efficiency

The portfolio's 6x ROAS validates the current investment level. However, the wide acquisition cost range (\$5,000–\$20,000) and the presence of a 22% Low ROAS campaign tier indicate that the portfolio contains pockets of highly inefficient spend. The business is currently treating all campaigns as equal investment priorities, a mistake that is compressing overall portfolio returns.

TechCorp's dominance of the bottom 10 campaigns is a direct budget efficiency risk. If TechCorp-associated campaigns represent even 5% of total spend (\$120M+), the business is allocating significant capital to campaigns generating only 2x ROAS while the portfolio average is 6x. This spread represents material opportunity cost.

5.2 Revenue Growth Requires Strategy, Not Just Spend

The flat daily revenue trend (\$40M–\$44M throughout 2021) is the most important strategic signal in this dashboard. At the current 6x ROAS, the portfolio is efficiently converting every dollar of ad spend into six dollars of revenue but efficiency alone is not growth. The revenue ceiling is structural, not accidental, and will not be broken by simply increasing spend at the same mix and strategy.

Breakthrough growth will require one or more of: (a) expanding addressable audience through new targeting segments or geographies, (b) introducing higher-value product or service lines into the conversion funnel, (c) exploiting seasonal amplification windows that the data confirms are currently untapped, or (d) differentiated creative strategies that push above the 14.04% CTR plateau.

5.3 Audience Fatigue & Creative Lifecycle

The CTR saturation pattern confirmed independently by both the Power BI dashboards and the Python EDA has direct financial consequences. Campaigns running beyond 5,000 impressions are paying for ad inventory that generates less than 12% CTR. At scale across 69,669 campaigns, this inefficiency compounds into millions of dollars of underperforming media spend.

The high variance in Engagement Score (std = 2.87) reinforces that creative quality is the differentiating variable. Campaigns with Engagement Score 1–3 are almost certainly contributing disproportionately to the low-CTR, high-impression tail. Creative rotation and quality thresholds should be operationalised as standard campaign management practice.

5.4 Customer Segment Strategy

The near-equal revenue distribution across four segments (\$2.88bn–\$2.93bn) is operationally reassuring, it confirms that the business has achieved genuine multi-segment reach. However, it also reveals that the marketing team has not yet identified a premium audience segment to over-index on. Revenue parity does not imply profit parity: if one segment has a 20% higher average order value or a 30% higher repeat purchase rate, equal investment allocation is suboptimal.

5.5 The Balanced Channel Mix Is a Resilience Strength

The near-perfect revenue split (16.48%–16.81% per channel) means the portfolio is not dependent on any single platform. This is a genuine competitive advantage in an environment of rising platform CPMs, algorithm changes, and audience fragmentation. If Instagram's costs increase by 20%, the business can absorb this by reallocating \$500M of spend to Email, Google Ads, or Website without materially impacting total returns.

5.6 Company & Vendor Management

The EDA finding that Innovate Industries accounts for 5 top-performing campaigns and TechCorp accounts for 5 worst-performing campaigns introduces a vendor management dimension not visible from the Power BI dashboards alone. If these companies represent external agencies, media vendors, or internal business units, the performance divergence warrants a formal performance review and contract/incentive restructuring.

6. Strategic Recommendations

Eight recommendations are presented in priority order. Each recommendation is directly traceable to the data findings from the Power BI dashboards and Python EDA.

PRIORITY LEGEND

● Critical — Immediate action | ● High — Within 30 days | ○ Medium — Within 90 days | ○ Low — Next planning cycle

#	Priority	Recommendation & Rationale
1	CRITICAL	Audit Campaign 170866 Immediately Analyse revenue contribution, customer lifetime value, and conversion quality for Campaign 170866. Its CAC of ~\$3,200 is 25% above the next most expensive campaign. If LTV does not justify the premium, pause and reallocate budget to Campaigns 153568, 138026, 190323, or 141396 all of which have CAC below \$2,000. Source: Dashboard 1 & 4 (CAC chart).
2	CRITICAL	Conduct Immediate TechCorp Performance Review TechCorp accounts for 5 of the 10 worst campaigns by ROI, including 3 campaigns with acquisition costs above \$17,000 generating only 2x returns. Initiate a formal performance review covering targeting strategy, creative quality, and channel selection. Establish a 90-day improvement threshold before considering budget suspension. Source: Python EDA (bottom 10 table).
3	HIGH	Implement Impression Frequency Caps and Creative Rotation Set a maximum of 3,000–5,000 impressions per audience segment across all campaigns. Introduce mandatory creative rotation every 2–3 weeks for campaigns that reach this ceiling. The CTR saturation curve confirms that impressions beyond this threshold generate diminishing returns, the business is paying for exposures that convert at less than 12%. Source: Campaign Efficiency Dashboards (CTR vs Impressions chart).
4	HIGH	Build a Seasonal Campaign Amplification Calendar The flat conversion rate and revenue trend across all 12 months confirms that seasonal demand signals are not currently being exploited. Build a demand calendar overlaying external events (holiday seasons, category peaks, cultural moments) with planned spend uplift windows. Test a 20–25% spend increase during 2 identified peak months in 2022 and measure whether revenue ceiling can be broken above \$44M/day.
5	HIGH	Commission a Customer Lifetime Value (LTV) Segmentation Study The near-equal revenue distribution across Fashionistas, Foodies, Health & Wellness, and Outdoor Adventurers does not reveal margin, repeat purchase, or LTV differences. Commission a LTV analysis and restructure budget weights to reflect true segment profitability, not just conversion volume. Even a 10% uplift in the highest-LTV segment would add \$288M+ in revenue. Source: Profitability Analysis Dashboard (segment table) + Python EDA.
6	MEDIUM	Introduce a Minimum ROAS Threshold and Quarterly Portfolio Review Establish a minimum acceptable ROAS of 3.0x for all active campaigns. The ~45,000 campaigns currently in the Low ROAS tier represent roughly 22% of the portfolio. Review and pause non-compliant campaigns quarterly, redirecting freed budget to Medium and High tier campaigns. A structured ROAS floor prevents gradual portfolio dilution. Source: Profitability Analysis Dashboard (ROAS category chart).
	MEDIUM	Test Differentiated Channel Budget Weights

7	MEDIUM	The current near-equal channel split (16.48%–16.81%) may not reflect each channel's true marginal return potential. Run a 90-day controlled experiment: increase Email and Google Ads budgets by 15% each (the highest ROI channels per the Python EDA) and reduce Instagram and Facebook proportionately. Measure whether the revenue ceiling shifts upward or whether equal weighting represents true market efficiency. Lever Innovate Industries Playbook Across Portfolio Innovate Industries holds 5 of the top 10 campaigns across Influencer and Email formats on multiple channels suggesting a repeatable, channel-agnostic performance approach. Commission a best-practice audit of Innovate Industries' targeting, creative strategy, and audience selection, and codify findings into a portfolio-wide playbook. Source: Python EDA (top 10 table).
8	MEDIUM	Test Seasonal Campaign Amplification Windows Despite stable Q1–Q4 CTR across all channels, external seasonal demand cycles (holiday periods, category events) may produce revenue uplifts not captured in engagement metrics. Overlay revenue-per-quarter data with external demand calendars and test a 15–20% budget increase during identified peak windows over the next 12 months. Source: Dashboard 1 & 4 (quarterly CTR chart).
9	LOW	Set Engagement Score Quality Thresholds The EDA revealed that Engagement Score has the highest variance of any metric (std = 2.87). Campaigns scoring 1–3 are likely contributing to the low-CTR, high-impression underperformance tail. Introduce a minimum Engagement Score threshold of 5 as a campaign activation requirement, and build a creative quality review gate for any campaign falling below this benchmark. Source: Python EDA (descriptive statistics).

7. Conclusion

The 2021 marketing analytics data presents a portfolio that is performing strongly at the macro level, \$14.5bn in revenue from a \$2.4bn investment, with consistent engagement across channels, campaign types, and customer segments. These are results any marketing organisation would be proud of.

However, the combined analysis of Power BI dashboards and Python EDA has revealed that this strong overall performance masks important pockets of inefficiency. TechCorp's systematic underperformance, Campaign 170866's acquisition cost outlier, the CTR saturation effect at high impression volumes, and the 22% Low ROAS campaign tail collectively represent hundreds of millions of dollars in addressable optimisation opportunity.

The headline finding from the first dasgis both reassuring and challenging: the portfolio is stable, efficient, and consistent but it is not growing. Revenue has plateaued within a \$40M–\$44M daily band, conversion rates are fixed between 5–8%, and all six channels contribute exactly one-sixth of total revenue every single month. This consistency is a tribute to strong operational execution. It is also a signal that the current strategy has reached its natural ceiling. The headline finding is both reassuring and challenging: the portfolio is stable, efficient, and consistent but it is not growing. Revenue has plateaued within a \$40M–\$44M daily band, conversion rates are fixed between 5–8%, and all six channels contribute exactly one-sixth of total revenue every single month. This consistency is a tribute to strong operational execution. It is also a signal that the current strategy has reached its natural ceiling

The path to a best-in-class portfolio is not to scale current spend uniformly, it is to eliminate the low-ROI tail, replicate the Innovate Industries playbook, apply impression frequency caps to combat creative fatigue, and develop a lifetime value-based segmentation model to concentrate resources where true profitability is highest.

It is recommended that this analysis be refreshed on a quarterly basis, with campaign-level revenue attribution data incorporated to enable more precise ROAS and ROI measurement at the individual campaign and segment level.